Cutnell & Johnson
Position, Velocity, and Acceleration
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ The purpose of this exploration is to study the relationships between position, velocity, and acceleration in linear motion and, in turn, how the relationships can be used to describe an object's motion.

§ Use the Motion Sensor to measure the motion of a cart as it moves away from the sensor.

§ Analyze the position and velocity of the cart on seperate graphs. [The Analysis section has instructions for determining the acceleration of an object from its position vs. time and velocity vs. time graphs. Pay close attention to the appearance of the velocity graph (linear) and the position graph (quadratic).]

Learning Outcomes

You will be able to:

- Measure the position and velocity of an object that has uniform acceleration.
- Explore the relationships between position, velocity, and acceleration.
- Analyze the graphs of position and velocity vs. time for a cart's motion.
- Explain how the slope of the position graph relates to the velocity graph.
- Recognize the general shape of position and velocity graphs for motion involving constant acceleration.

Hypothesis

§ What is the shape of the position versus time graph for an object with uniform acceleration?

§ What is the shape of the velocity versus time graph for an object with uniform acceleration?

Background

§ The equations of translational (or linear) kinematics can be used for any moving object, as long as the acceleration of the object is constant. The equations to the right can be used to solve problems when constant acceleration is present. Later in this exploration you will use these equations to answer some questions about the data you will record.

$$v = v_0 + at$$

$$x = \frac{1}{2}(v_0 + v)$$

$$x = v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2ax$$

» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 2, Section 2.4.

Materials

- PASPORT USB Link
- PASPORT Motion Sensor
- Fan Accessory
- PAScar
- 1.2 m Dynamics Track

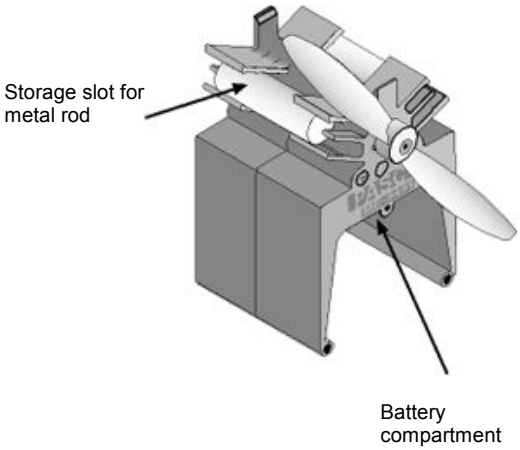
Setup *Equipment Setup*

Safety Reminder: Follow the directions for using the equipment.

§ Put batteries in two of the slots in the base of the Fan Accessory and put the metal rods in the other two slots. (Two batteries makes the fan run at low speed.)

§ Attach the Fan Accessory to the cart and place them on the track. Level the track so the cart will not roll on its own (you could use a bubble level to make sure).

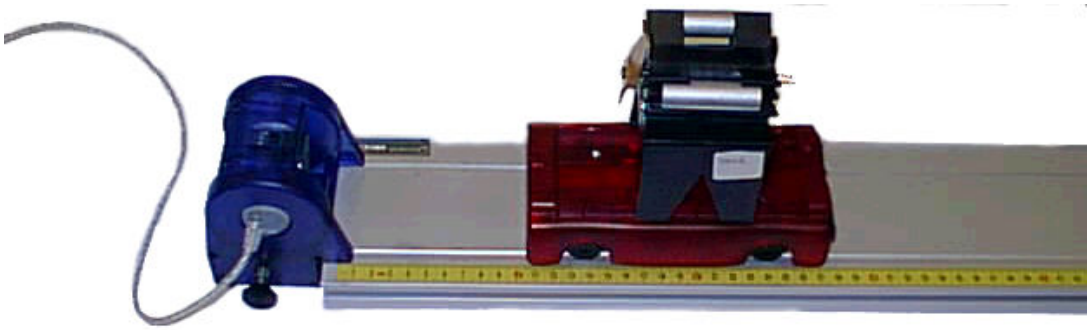
§ Attach the Motion Sensor to the end of the track and make sure there is nothing to obstruct the signal going from the front of the Motion Sensor to the cart. Make sure the fan will push the cart away from the Motion Sensor.



Setup *Equipment Setup*

§ Hold the cart about 15 cm in front of the Motion Sensor.

§ Start the Fan Accessory (make sure the fan will push the cart away from the sensor.)



» **Note:** Your setup may vary.

Process *Record Data*

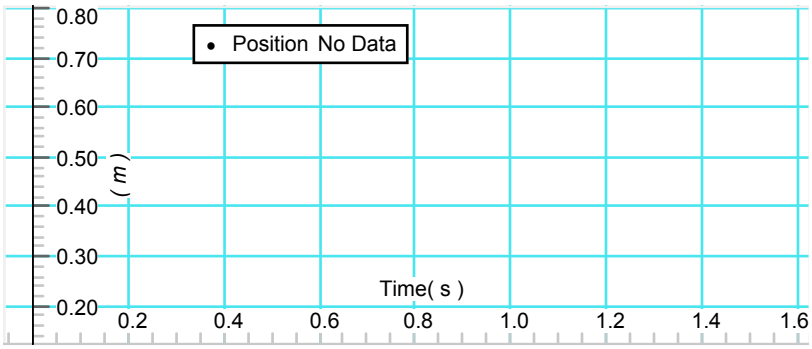
§ Click **Start** to begin recording data.

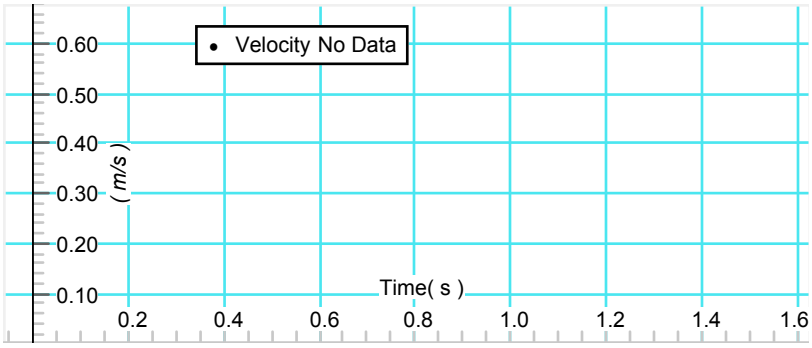
§ Release the cart so the fan can push the cart away from the Motion Sensor.

§ Click **Stop** when the cart has nearly reached the end of the track.

§ Turn off the Fan Accessory.

§ Click the **Scale to fit** button if necessary.





Observations

§ What is the shape of the position versus time graph?

§ What is the shape of the velocity versus time graph?

§ How well did your predictions match the motion sensor generated plots?

Analyze *Variables and Controls*

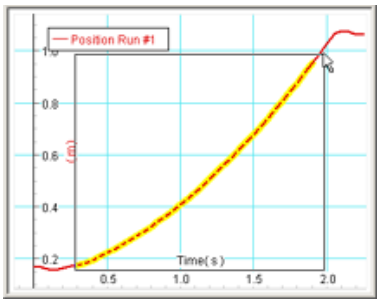
§ What were the variables in this activity?

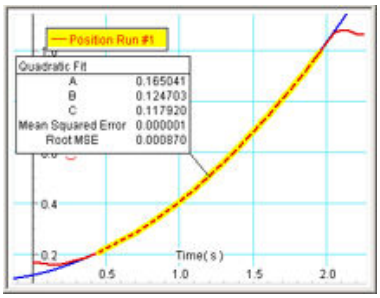
§ Which of those did you control?

§ Which did you manipulate and what was the response?

Analyze *How to Find the Acceleration from Position*

This page describes how to use the analysis tools in the graph display to find the acceleration from the position versus time graph.





Images of example data

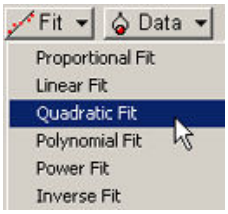
§ First Step: Highlight a smooth part of the position graph.

» If you don't see a very smooth section, zoom in or out to try to find it. If that doesn't work, try taking another set of data.

§ Second Step: Click the **Fit** menu and select **Quadratic Fit**.

» The "**A**" is the coefficient of the squared term in the curve fit formula.

Fit menu



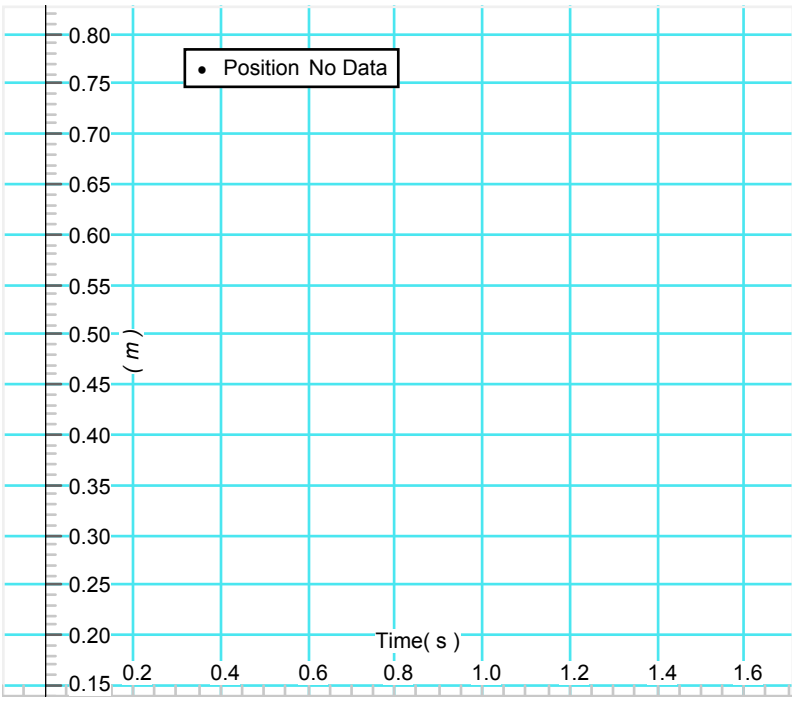
Analyze *Find the Acceleration from Position versus Time*

§ Highlight a smooth part of the position graph.

§ Click the **Fit** menu and select **Quadratic Fit**.

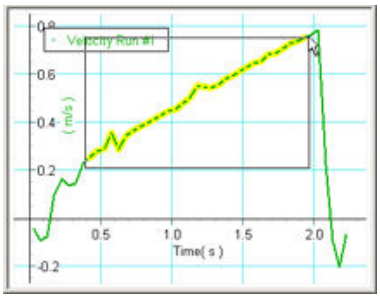
§ The "**A**" is the coefficient of the squared term in the curve fit formula.

A =



Analyze *How to Find the Acceleration from Velocity*

This page describes how to use the analysis tools in the graph display to find the acceleration from the velocity versus time.



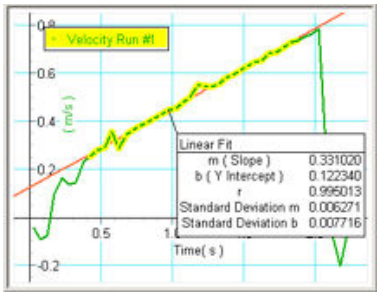


Image of example data

§ First Step: Highlight a smooth part of the velocity graph.

» If you don't see a very smooth section, zoom in or out to try to find it. If that doesn't work, try taking another set of data.

§ Second Step: Click the **Fit** menu and select **Linear Fit**.

» The "**m**" value is the slope of the line that is the best fit for the data. The slope of velocity versus time is the acceleration.

Fit menu

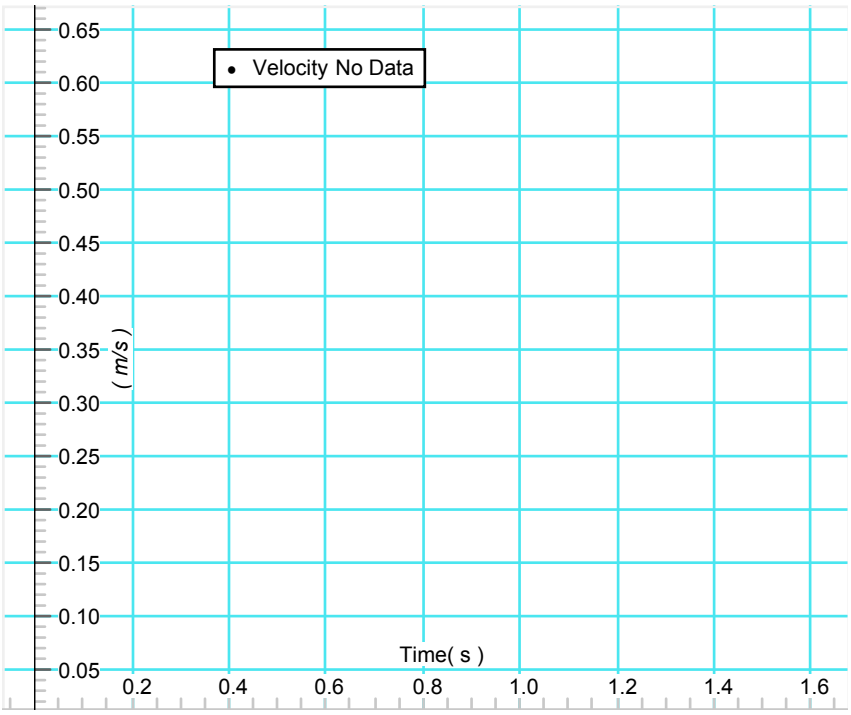
Analyze *Find the Acceleration from Velocity and Time*

§ Highlight a smooth part of the velocity graph.

§ Click the **Fit** menu and select **Linear Fit**.

§ The "**m**" value is the slope. Record the slope as the acceleration.

m =



Analyze *Data Analysis and Calculations*

» Here is the value of the "**A**" coefficient from the **Quadratic Fit** for the position versus time graph.

§ Multiply this number by "2" to get the *acceleration* based on position versus time. Put the value of 2 x "**A**" into the box.

» Here is the value of the slope of the **Linear Fit** for the velocity versus time graph. This is the *acceleration* based on velocity versus time.

§ Using the acceleration from your position versus time data and the equations of motion from the Background section, calculate the time it would take for the PAScar to travel 50 meters.

§ Using the time you found in the previous problem, calculate how fast the PAScar would be traveling.

$$v = v_0 + at$$

$$x = \frac{1}{2}(v_0 + v)$$

$$x = v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2ax$$

»Time:

»Speed:

Synthesize *Verification*

§ Compute the percent difference (% Difference) of your two values for acceleration.

$$\% \text{ Difference} = \left| \frac{\text{acceleration from position - acceleration from velocity}}{\text{acceleration from velocity}} \right| \text{x100}$$

»Acceleration from position versus time (2 x "**A**" value)

»Acceleration from velocity versus time (slope)

»Percent difference (% Difference):

Synthesize *Conclusions and Error Analysis*

§ In this exploration you used a graphical representation to show how position and velocity are related for an object with constant acceleration. What did you discover?

§ What were the sources of error in this experiment?

Synthesize *Applications and Extension Problem*

§ What are some examples of where the equations of motion are used?

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 2, Concept Question 85:

(a) The acceleration of a NASCAR race car is zero. Does this necessarily mean that the velocity of the car is also zero?
(b) If the speed of the car is constant as it goes around the track, is its average acceleration necessarily zero? Justify your answers.